

Highlights

Model-Guided Design of Porous Polyimide/Boron Nitride Films for Low-Permittivity, Thermally Conductive Packaging

- Porous PI/h-BN films made by NIPS over the full 0–70 wt% h-BN range
- Three-phase Lewis–Nielsen model unifies dielectric and thermal behavior
- Pore shape factor $A = 0.08$ fingerprints NIPS sponge vs. spherical pores
- Single-parameter thermal fit: $\lambda_{\text{BN,eff}} = 6.8 \text{ W m}^{-1} \text{ K}^{-1}$, interface-limited
- 70 wt% h-BN gives the best measured ε_r – λ balance; model loci agree